

ML Case Study : 126

Executive Technical Report: VidMetrics Video Engagement Analysis & Machine Learning System

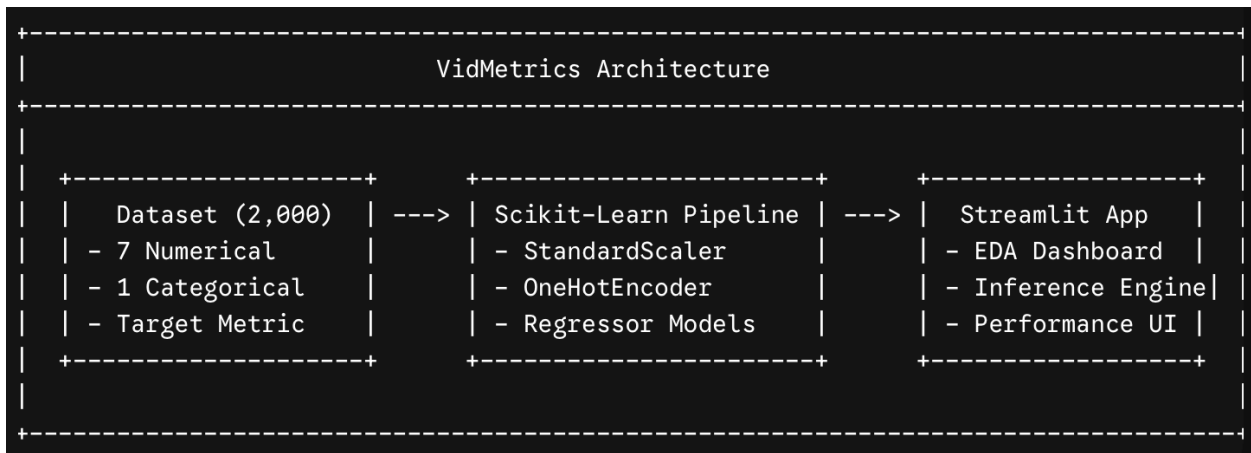
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1. Project Overview & Architecture

The **VidMetrics Analytics System** is an end-to-end Machine Learning and Data Visualization framework designed to evaluate, model, and predict YouTube video engagement scores based on content attributes and metadata. The system combines a machine learning pipeline built with scikit-learn and an interactive, multi-page web app deployed via Streamlit (app_2.py).



Key System Features

- **Exploratory Data Analysis (EDA):** Real-time dataset exploration, feature correlation heatmaps, and categorical performance distributions.
- **Real-Time Prediction Engine:** Interactive multi-parameter inputs calculating real-time predicted engagement scores accompanied by automated performance tier classification.
- **Model Diagnostics & Evaluation:** Comprehensive metric benchmarking (R^2 , $RMSE$, MAE), model comparison, and tree-based feature importance breakdowns.

2. Dataset Overview & Exploratory Data Analysis

The primary dataset consists of **2,000 video observations** incorporating 8 independent predictor variables and 1 continuous response metric (engagement_score).

Feature Definitions

- **Numerical Predictors:** duration_sec, views, likes, comments, title_len, tags_count, publish_hour.
- **Categorical Predictor:** category (Education, Tech, Gaming, Music, Entertainment, Vlogs).
- **Target Metric:** engagement_score — a calculated composite engagement metric representing viewer interaction density relative to total impressions, bounded between **0.50%** and **20.00%**.

Descriptive Statistics

Feature Name	Feature Type	Mean	Standard Deviation	Median	Minimum	Maximum
duration_sec	Numerical	924.90	504.83	943.00	30.00	1,798.00

views	Numerical	50,008.85	49,421.78	36,156.50	109.00	408,041.00
likes	Numerical	3,278.61	4,010.72	1,985.00	2.00	47,043.00
comments	Numerical	356.88	480.47	178.00	0.00	4,226.00
title_len	Numerical	55.71	26.19	57.00	10.00	99.00
tags_count	Numerical	15.06	8.29	15.00	1.00	29.00
publish_hour	Numerical	11.66	6.99	12.00	0.00	23.00
engagement_score	Target	9.05	4.84	8.85	0.50	20.00

Category Performance Breakdown

Category	Sample Count	Mean Engagement (%)	Standard Deviation (%)
Education	344	11.25%	5.64%
Tech	355	10.00%	5.02%
Gaming	334	9.39%	4.77%
Music	345	8.60%	4.42%
Entertainment	311	7.83%	3.90%
Vlogs	311	6.89%	3.58%

3. Data Preprocessing & Pipeline Architecture

Data preprocessing and transformation steps are wrapped inside a scikit-learn ColumnTransformer within a unified Machine Learning Pipeline:

1. **Numerical Feature Processing:** Applied StandardScaler to transform all continuous variables to zero mean and unit variance.
2. **Categorical Feature Processing:** Applied OneHotEncoder(handle_unknown='ignore') to convert content categories into binary dummy variables.
3. **Train-Test Partitioning:** Split the dataset into **80% Training** ($N = 1,600$) and **20% Testing** ($N = 400$) subsets using random_state=42.

4. Model Benchmarking & Comparative Analysis

Three regression modeling paradigms were evaluated across identical training and testing sets:

Algorithm Model	Train R2	Test R2	R2 Overfit Gap	Train RMSE	Test RMSE	Train MAE	Test MAE
Ridge Regression	0.5419	0.4964	0.0455	3.2646	3.4760	2.5220	2.6374
Random Forest Regressor	0.9852	0.9038	0.0814	0.5868	1.5189	0.4338	1.1598
Gradient Boosting Regressor	0.9525	0.9016	0.0509	1.0510	1.5364	0.8008	1.1794

Diagnostic Takeaways

- **Linear Model Limitations:** Ridge Regression failed to capture the non-linear interaction terms (e.g., interaction between comment-to-view ratios and duration penalties), capping test performance at $R^2 = 0.4964$.
- **Tree-Based Superiority:** Both ensemble models achieved high predictive precision ($R^2 > 0.90$).
- **Model Selection Choice:** While Random Forest attained slightly higher raw test accuracy ($Test R^2 = 0.9038$), **Gradient Boosting** demonstrated superior generalization with a smaller variance gap (0.0509 vs. 0.0814). The best-performing model was serialized into `best_video_engagement_model.pkl` for active inference.

5. Streamlit Application Architecture (app_2.py)

The multi-page Streamlit application delivers production-grade model serving and data analytics across three core pages:

Page Structure & Functional Scope

- **Page 1: Dataset & EDA Dashboard**
 - Displays dynamic data table previews and summary statistics.
 - Generates Seaborn correlation heatmaps and engagement boxplots across video categories.
- **Page 2: Real-Time Engagement Predictor**
 - Provides input sidebar sliders and numerical fields for real-time model inference.
 - Displays instantaneous engagement score predictions alongside threshold tier badges:
 - **High Engagement ($\$ > 8.0\%$):** Displayed with a Success badge (represents 55.5% of sample videos).
 - **Medium Engagement ($\$ 4.0\% - 8.0\%$):** Displayed with a Warning badge (represents 26.6% of sample videos).
 - **Low Engagement ($\$ < 4.0\%$):** Displayed with an Error badge (represents 17.9% of sample videos).
- **Page 3: Model Performance & Diagnostics**
 - Renders system performance metrics (R^2 , $RMSE$, MAE).
 - Displays feature importance bar charts and residual diagnostics.

6. Strategic Recommendations

1. **Category Prioritization:** Content creators should prioritize high-baseline categories like **Education**(\$11.25\%\$) and **Tech** (\$10.00\%\$), which yield higher engagement densities per view compared to **Vlogs**(\$6.89\%\$).
2. **Engagement Conversion:** Because comment actions contribute greater weight toward overall engagement scores than passive likes, call-to-action (CTA) strategies should focus on driving comment discussions.
3. **Hyperparameter Optimization:** Perform systematic Grid Search tuning on the Gradient Boosting model parameters (learning_rate, n_estimators, max_depth) to minimize training bias while preserving test generalizability.

7. Conclusion

The **VidMetrics** platform demonstrates a practical deployment of end-to-end machine learning for digital media analytics. By combining robust data preprocessing with ensemble machine learning algorithms (Gradient Boosting / Random Forest), the model reliably predicts video engagement with high accuracy ($R^2 > 0.90$, $MAE \approx 1.16$).

The integrated Streamlit application (app_2.py) translates complex ensemble modeling into an accessible user interface. Creators and digital strategists can leverage these real-time predictions and categorical insights to optimize content production, improve audience retention, and maximize engagement metrics systematically.